state to decay. There are a few nuclides having more than one metastable state. These are then designated as $m_1$, $m_2$, etc.

## 3.2. ATOMIC MASSES AND ATOMIC WEIGHTS

The *universal mass unit*, abbreviated u (sometimes amu for atomic mass unit), is defined as one-twelfth of the mass of the $^{12}C$ atom which has been defined to be exactly 12 u. The absolute mass of a $^{12}C$ atom is obtained by dividing the value 12 by the Avogadro number ($\mathbf{N}_A = 6.022137 \times 10^{23}$). The value for the mass of a $^{12}C$ atom, i.e. the nucleus plus the 6 extranuclear electrons, is thus $1.992\,648 \times 10^{-23}$ g. Atomic masses are expressed in units of u relative to the $^{12}C$ standard. This text uses $M$ to indicate masses in units of u, and $m$ in units of kilograms; $m = M/10^3\ \mathbf{N}_A$. One atomic mass unit corresponds to 931.5 MeV using the famous equation $E = m_0c^2$. Historically this has not always been the case. Up to as late as the 1960ies two different scales of atomic masses existed. The chemists used a scale based on the natural composition of oxygen and designated that to have a mass of 16 u while in the physics community the same mass was assigned to the most common oxygen isotope, $^{16}O$. Since natural oxygen also contains the isotopes $^{17}O$ and $^{18}O$ this led to two different tables of atomic masses which naturally was highly inconvenient.

In nuclear science it has been found convenient to use the atomic masses rather than nuclear masses. The number of electrons are always balanced in a nuclear reaction, and the changes in the binding energy of the electrons in different atoms are insignificant within the degree of accuracy used in the mass calculations. Therefore the difference in atomic masses of reactants and products in a nuclear reaction gives the difference in the masses of the nuclei involved. In the next chapter, where the equivalence between mass and energy is discussed, it is shown that all nuclear reactions are accompanied by changes in nuclear masses.

The mass of the nucleus can be approximated by subtracting the sum of the masses of the electrons from the atomic mass. The mass of an electron is 0.000549 u which is equivalent to $9.1094 \times 10^{-31}$ kg. Since the neutral carbon atom has 6 electrons, the approximate mass of the nucleus is $1.992\,648 \times 10^{-26} - 6 \times (9.1094 \times 10^{-31}) = 1.992\,101 \times 10^{-26}$ kg. This calculation has not included the difference in the mass of the 6 extra electrons attributable to the binding energy of these electrons. However, this binding energy has a mass equivalence which is smaller than the least significant figure in the calculation.

The mass of a neutron is 1.008 665 u while that of the hydrogen atom is 1.007 825 u. This shows the fact that the neutron decays to a proton and en electron. Since both neutrons and protons have almost unit atomic masses, the atomic mass of a nuclide should be close to the number of nucleons, i.e. the mass number. However, when the table of elements in the periodic system (Appendix I) is studied it becomes obvious that many *elements* have masses which are far removed from integral values. Chlorine, for example, has an atomic mass value of 35.453 u, while copper has one of 63.54 u. The variations in precision of the given atomic weights can be explained by the effect of the relative abundances of the stable isotopes of the elements contributing to produce the observed net mass.

If an element consists of $n_1$ atoms of isotope 1, $n_2$ atoms of isotope 2, etc., the *atomic fraction* $x_1$ for isotope 1 is defined as:

$$x_1 = n_1/(n_1 + n_2 + \ldots) = n_1/\Sigma n_i \tag{3.3}$$

The *isotopic ratio* is the ratio between the atomic fractions (or *abundances*) of the isotopes. For isotopes 1 and 2, the isotopic ratio is

$$\zeta_1 = x_1/x_2 = n_1/n_2; \;\; \zeta_2 = x_2/x_1 = n_2/n_1 \tag{3.4}$$

The *atomic mass of an element* (or *atomic weight*) $M$ is defined as the average of the isotopic masses, i.e. $M_i$ is weighted by the atomic fraction $x_i$ of its isotope:

$$M = x_1\, M_1 + x_2\, M_2 + \ldots = \Sigma x_i\, M_i \tag{3.5}$$

As an example, natural chlorine consists of two isotopes of which one has an abundance of 75.77% and an atomic mass of 34.9689 u and the second has an abundance of 24.23% and a mass of 36.9659 u. The resultant average atomic mass for the element is 35.453 u. The atomic mass of copper of 63.546 u can be attributed to the presence of an isotope in 69.17%

**Table 3.1** Isotopic data for some elements

| Element | *Z* | *N* | *A* | Atomic mass (u) | Abundance (%) | Atomic weight | Symbol |
|---|---|---|---|---|---|---|---|
| Hydrogen | 1 | 0 | 1 | 1.007 825 | 99.985 | | $^{1}$H |
| | 1 | 1 | 2 | 2.014 102 | 0.0155 | 1.00797 | $^{2}$H, D |
| | 1 | 2 | 3 | 3.016 049 | 0 | | $^{3}$H, T |
| Helium | 2 | 1 | 3 | 3.016 030 | 0.0001 | | $^{3}$He |
| | 2 | 2 | 4 | 4.002 603 | 100.00 | 4.0026 | $^{4}$He |
| Lithium | 3 | 3 | 6 | 6.015 121 | 7.42 | 6.939 | $^{6}$Li |
| | 3 | 4 | 7 | 7.016 003 | 92.58 | | $^{7}$Li |
| Beryllium | 4 | 5 | 9 | 9.012 182 | 100.00 | 9.0122 | $^{9}$Be |
| Boron | 5 | 5 | 10 | 10.012 937 | 19.6 | 10.811 | $^{10}$B |
| | 5 | 6 | 11 | 11.009 305 | 80.4 | | $^{11}$B |
| Carbon | 6 | 6 | 12 | 12.000 000 | 98.892 | 12.0112 | $^{12}$C |
| | 6 | 7 | 13 | 13.003 355 | 1.108 | | $^{13}$C |
| Nitrogen | 7 | 7 | 14 | 14.003 074 | 99.635 | 14.007 | $^{14}$N |
| | 7 | 8 | 15 | 15.000 109 | 0.365 | | $^{15}$N |
| Oxygen | 8 | 8 | 16 | 15.994 915 | 99.759 | | $^{16}$O |
| | 8 | 9 | 17 | 16.999 131 | 0.037 | 15.999 | $^{17}$O |
| | 8 | 10 | 18 | 17.999 160 | 0.204 | | $^{18}$O |
| Chlorine | 17 | 18 | 35 | 34.968 853 | 75.8 | 35.453 | $^{35}$Cl |
| | 17 | 20 | 37 | 36.965 903 | 24.2 | | $^{37}$Cl |
| Uranium | 92 | 143 | 235 | 235.043 924 | 0.724 | 238.029 | $^{235}$U |
| | 92 | 146 | 238 | 238.050 785 | 99.266 | | $^{238}$U |

abundance with a mass of 62.9296 u and of a second isotope of 30.83% abundance and 64.9278 u. Atomic masses and abundances of some isotopes are given in Table 3.1.

## 3.3. DETERMINATION OF ISOTOPIC MASSES AND ABUNDANCES

### 3.3.1. The mass spectrometer

The masses and relative abundances of different isotopes occurring naturally in an element can be determined with great exactness using the same technique J. J. Thomson employed to demonstrate the presence of isotopes in neon. The instrument used for this purpose is known as a mass spectrometer. The principles of the electromagnetic *mass spectrometer* are shown in Figs. 3.1 and 3.2.

Let us first consider the movement of an ion in electric and magnetic fields, as shown in Fig. 3.1. The ion of mass $m$ is assumed to have a charge $q$ (coulomb), which is an integer ($z$) multiple of the elementary charge **e** ($1.602\ 177 \times 10^{-19}$ C): $q = z\mathbf{e}$. If it is accelerated from velocity zero to $v$ (m $s^{-1}$) by an electric potential $V$ (volts), it acquires a kinetic energy $E_{kin}$ corresponding to joule (or Newton meter).

$$E_{kin} = \tfrac{1}{2}mv^2 = qV \tag{3.6}$$

If $q$ is given in units of the elementary charge, the kinetic energy is in units of *electron volts* (eV). For transformation to other energy units, see Appendix IV.

Figure 3.1 shows the deviations of a positive ion in an electric field $U$ (newton/coulomb) directed upwards, and a magnetic field $B$ (tesla) directed into the page. The force $F$ (newton) acting on the ion due to the electric field is

$$F_e = qU \tag{3.7}$$

If the electric force would dominate, the ion would hit the screen at point $P_1$. For the magnetic field only, the force is

$$F_m = qvB \tag{3.8}$$

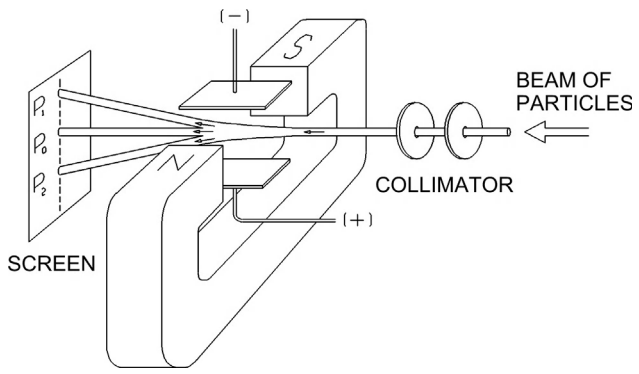



**Figure 3.1** Movement of positive ions in electric and magnetic fields.